

Multidisciplinary Tumor Board

Encounter Type: Hematology day-unit visit & MDT discussion

Date: 16.08.2023

Patient: Linda Taylor (14.02.1952) – MRN AML010

Primary Dx: AML, **ELN 2022 Intermediate**, **FLT3-ITD** positive on 11.10.2022

First-line therapy: **Gilteritinib** (single-agent) initiated **20.10.2022** based on FLT3 status and comorbidity profile precluding intensive chemotherapy at diagnosis. Initial hematologic improvement noted with rising ANC and platelet recovery.

Interim Course:

- Nov–Dec 2022: Cytopenias manageable; no significant transaminitis; QTc remained <470 ms.
- Jan–Apr 2023: Clinical remission by peripheral counts; transfusion independence achieved; marrow in April showed low-level residual disease (blasts 3–4%), FLT3-ITD still detectable at low VAF.
- **July 2023:** Gradual fall in counts with rising LDH and circulating blasts. **Bone marrow 21.07.2023:** 36% blasts, persistent **FLT3-ITD**, evolving cytogenetics with +8. Findings consistent with **relapsed/refractory disease while on gilteritinib**.

Comorbidities: Hypertension, stage-3a CKD (Cr 1.3–1.5), hypothyroidism, prior DVT (2019, completed anticoagulation), osteopenia. ECOG 1–2.

Current Symptoms/Exam: Fatigue, easy bruising, no fevers. Spleen tip palpable 1–2 cm below LCM. ECOG 2 today. No active bleeding.

Pertinent Labs (24.08.2023): WBC $18.2 \times 10^9/L$ with 28% blasts (peripheral), Hgb 8.4 g/dL, Plt $32 \times 10^9/L$; uric acid 6.9 mg/dL; potassium 4.8 mmol/L; creatinine 1.45 mg/dL (eGFR ~42 mL/min/1.73m²); AST/ALT 28/24; LDH 690 U/L; fibrinogen 420 mg/dL; D-dimer elevated.

Imaging: Portable CXR clear. Baseline echo earlier in 2023 with LVEF 59%.

Assessment:

- **Relapse of FLT3-ITD AML on 29.07.2023.** She remains alive and is medically appropriate for salvage therapy with curative intent if a remission can be achieved, though renal function and age require careful dosing.

MDT Recommendations / Second-Line Strategy:

1. **Salvage regimen: FLAG-IDA + venetoclax (7 days)** with dose-attenuation for age/renal function **OR azacitidine + venetoclax + sorafenib** if intensive regimen is contraindicated. Given ECOG 2 and CKD, we favor **azacitidine (75 mg/m² D1-7) + venetoclax** with careful ramp (reduce dose if concomitant azoles are needed) and **sorafenib 200 mg bid** as an off-label FLT3 blockade switch.
 - **TLS prophylaxis** and inpatient monitoring during ramp due to circulating blasts and renal function.
 - **Antimicrobial prophylaxis:** levofloxacin, posaconazole (with venetoclax dose adjustment), and acyclovir; consider atovaquone for PJP if profound neutropenia persists.
 - Transfusion thresholds: Hgb ≥ 8 , platelets ≥ 10 (≥ 20 if bleeding/risk).
2. **Transplant planning:** If CR or CRi achieved, proceed to **allogeneic HSCT** using a reduced-intensity conditioning (RIC) regimen. HLA typing of siblings/registry already initiated; cardiopulmonary evaluation ordered.
3. **Supportive care:** Manage uric acid (allopurinol or rasburicase if needed), maintain aggressive hydration, correct electrolytes, and monitor QTc. Growth factor support post-cytopenic nadir as needed.
4. **Palliative integration:** Early palliative care consult for symptom management and to discuss goals of care alongside disease-directed therapy.
5. **Clinical trials:** If inadequate response to venetoclax-based salvage, consider enrollment in trials targeting FLT3 resistance or menin inhibitors if molecular evolution arises.

Discharge/Follow-up Plan (16.08.2023):

- Admit tomorrow for initiation of **azacitidine + venetoclax** with FLT3 switch inhibition as above.
- Continue home meds: amlodipine 5 mg daily, levothyroxine 50 mcg daily, vitamin D; **avoid NSAIDs** due to thrombocytopenia/renal function.
- Education provided regarding neutropenic precautions, fever threshold ($\geq 38.0^{\circ}\text{C}$), bleeding signs, and when to present emergently.
- Next marrow assessment planned after cycle 1 day ~28 to evaluate response and expedite transplant referral if eligible.

Summary Sentence: Ms. Taylor experienced **relapse on first-line gilteritinib**; we are moving forward with **second-line venetoclax-based salvage** with intent to bridge to **allogeneic transplantation** if remission is achieved.

Signed on behalf of the MDT,
[Name], MD — Hematology
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[Name], RN — BMT Coordinator